

T0 Theory and 't Hooft's Cellular Automaton Interpretation:

Deterministic Quantum Mechanics from Two Directions —
Where Both Agree, Where T0 Goes Deeper,
and Whether the Cellular Automaton Is a Special Case of T0

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- $G = \xi^2 \ell_p^2 c^3 / \hbar (= \xi^2 G) \rightarrow G = \ell_p^2 c^3 / \hbar = L_0^2 c^3 / (\xi^2 \hbar).$

Abstract

Gerard 't Hooft's *The Cellular Automaton Interpretation of Quantum Mechanics* (Springer, 2016) and T0 theory pursue the same fundamental goal: to understand quantum mechanics as deterministic and to replace the Copenhagen interpretation with a deeper classical mechanism. This document systematically analyses where both approaches agree (determinism, rejection of wavefunction collapse, discrete time structure at the Planck scale, Hilbert space as a tool) and where T0 is more fundamental: 't Hooft postulates the cellular automaton as a new foundation without explaining why precisely this structure. T0 derives the discrete spacetime structure from a single geometric parameter $\xi = 4/30\,000$. The central thesis of this document: 't Hooft's cellular automaton is not an independent postulate but a limiting case of T0 geometry at the sub-Planck scale $L_0 = \xi \cdot \ell_p$. It is further shown that 't Hooft's open problems (locality, fermions, gravity, free parameters of the Standard Model) receive natural answers in T0 through the time-mass duality $T \cdot m = 1$ and fractal spacetime geometry.

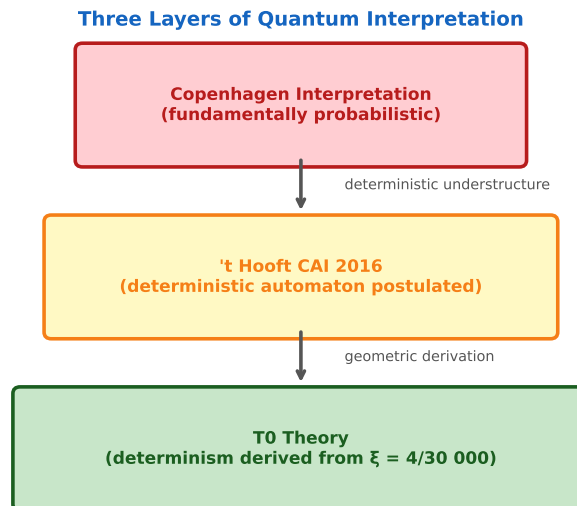


Figure 1: Three layers of quantum interpretation: Copenhagen (probabilistic), CAI by 't Hooft (automaton postulated), T0 theory (determinism derived from ξ). Each level subsumes the previous one.

.1 Introduction: Two Nobel-Prize Intuitions, One Goal

The Shared Discomfort

Gerard 't Hooft, Nobel Laureate 1999 for elucidating the quantum structure of electroweak interactions, has spent recent decades intensely occupied with a question most of his colleagues considered settled: is quantum mechanics fundamentally probabilistic, or does a deterministic structure underlie it?

His answer, laid out in detail in *The Cellular Automaton Interpretation of Quantum Mechanics* (2016), is unambiguous: quantum mechanics is a tool, not a theory. Beneath it lies a classical-deterministic mechanism operating at the Planck scale whose effective description yields familiar quantum mechanics.

T0 theory (time-mass duality, FFGFT) arrives at the same conclusion from a completely different direction. Starting from the observation that all physical constants emerge from a single geometric parameter $\xi = 4/30\,000$, not only does determinism follow as a consequence, but also a concrete geometric structure of the Planck scale that contains 't Hooft's cellular automaton as a special case.

Structure of This Document

1. **Section 2:** Core points of 't Hooft's CAI — what it claims, what it proves, what it leaves open.
2. **Section 3:** Six points of deep agreement between CAI and T0.
3. **Section 4:** Four fundamental differences — where T0 goes deeper.
4. **Section 5:** The central thesis: the cellular automaton as a limiting case of T0.
5. **Section 6:** 't Hooft's open problems and T0 answers.
6. **Section 7:** Testable predictions from the comparison.

.2 Core Points of the Cellular Automaton Interpretation

The Fundamental Thesis

't Hooft begins with a fundamental critique of the Copenhagen interpretation. The standard interpretation of quantum mechanics leaves several uncomfortable gaps: wavepacket collapse upon measurement, the non-locality of entanglement, the impossibility of a classical description of reality, and the apparent incompatibility with relativity theory at a fundamental level.

't Hooft's counter-proposal in one sentence: *quantum mechanics is the effective theory of a classical-deterministic system operating at the Planck scale as a cellular automaton.*

The Cogwheel Model as Prototype

The simplest model of the CAI is the *cogwheel model*: a system with N discrete states $|n\rangle$, $n = 0, \dots, N-1$, evolving according to a fixed rule: at each time step δt , one has $n \rightarrow n+1 \pmod{N}$. This is completely classical-deterministic.

't Hooft now shows that a Hilbert space and Hamiltonian operator can be assigned to this system:

$$H_{\text{op}}|n\rangle = \frac{2\pi k}{N\delta t}|n\rangle \quad (1)$$

where k is the energy quantum number. The states obey the Schrödinger equation — not because the system is quantum mechanical, but because quantum mechanics *is precisely this*: an effective description of periodic deterministic systems.

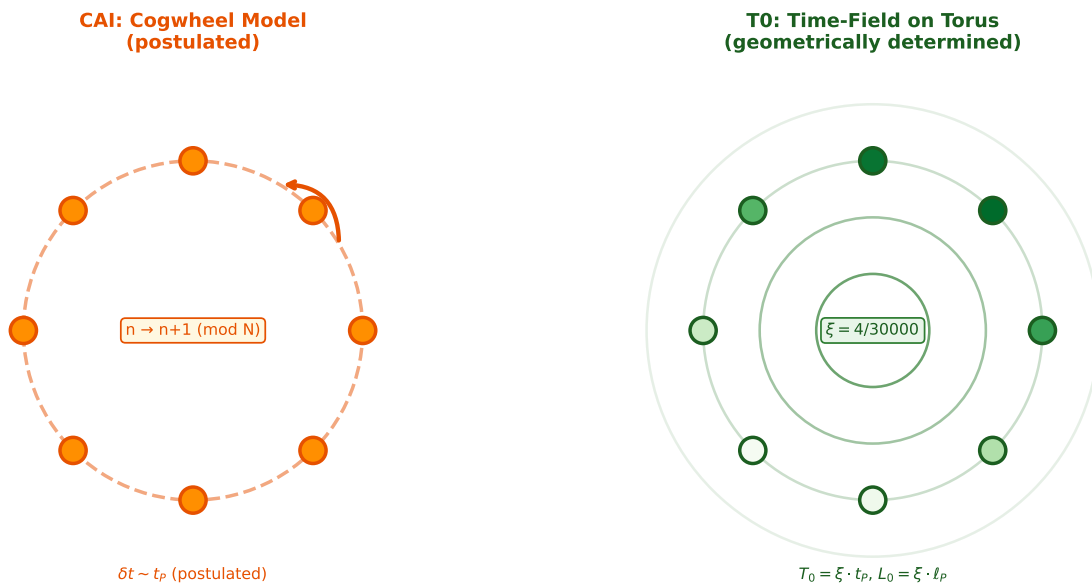


Figure 2: Left: CAI cogwheel with $N = 8$ discrete states (postulated). Right: T0 time field on geometric torus, lattice spacing $L_0 = \xi \cdot \ell_P$ (derived).

Ontological vs. Template States

A central concept of the CAI is the distinction between:

- **Ontological states** $|n\rangle_{\text{ont}}$: the actual states of the classical-deterministic system. At each moment, the system is in exactly one such state.

- **Template states** $|\psi\rangle$: superpositions of ontological states. These are not a picture of reality but description tools for an observer who does not know the exact initial conditions.

Probabilism in quantum mechanics is, according to 't Hooft, *epistemic*: it reflects the observer's ignorance of the precise ontological initial conditions, not a fundamental indeterminacy of nature.

Bell's Theorem and Superdeterminism

The apparently strongest argument against hidden-variable theories is Bell's theorem: no local, realistic theory can reproduce the quantum correlations. 't Hooft responds with the concept of *superdeterminism*: if the universe is completely deterministic, then even the settings of measurement apparatuses are determined by the initial conditions. The degrees of freedom assumed to be independent in Bell's argument are not.

't Hooft shows that this objection is in principle surmountable, but concedes that a complete quantitative theory of superdeterminism is still outstanding.

The Open Problems

't Hooft is explicitly transparent about the limits of his approach. At the end of the book he names explicitly:

- The **locality problem**: how are quantum correlations reproduced locally from the automaton?
- The **fermion problem**: how do half-integer spins arise from a boson-like automaton?
- The **gravity problem**: the automaton has not been connected to general relativity.
- The **free-parameter problem**: the Standard Model has 20+ free parameters. The automaton does not explain why they have these values.
- The **information-loss problem**: black holes and unitarity remain unresolved.

't Hooft's Programme

The CAI is a programme, not a finished theory. It shows that deterministic quantum mechanics is *possible*, but it does not yet provide a complete derivation of known physics from a deterministic foundation.

.3 Six Points of Deep Agreement

1. Determinism as a Fundamental Principle

Both approaches reject the fundamental probabilism of the Copenhagen interpretation. In the CAI, randomness is epistemic — it reflects ignorance of initial conditions. In T0, randomness is equally epistemic: the time field $T(x, t) = \hbar/(mc^2)$ evolves completely deterministically according to the T0 energy functional:

$$E_{T0} = \int |\nabla T(x)|^2 d^4x \quad (2)$$

Apparent quantum uncertainty in T0 is the projection of a deterministic four-dimensional time field onto the three-dimensional spatial hypersurface of an observer.

2. Rejection of Wavepacket Collapse

't Hooft: collapse is not a physical process but an information update of the observer about the ontological state of the system. Before the measurement, the system was in a definite state — the observer simply did not know it.

T0: identical conclusion from a different foundation. The time field $T(x, t)$ does not collapse — it evolves continuously. A measurement is an interaction of two time fields whose correlation generates the apparent collapse structure.

3. Hilbert Space as a Tool, Not Ontology

't Hooft writes explicitly: *quantum mechanics is viewed as a tool, not as a theory*. Hilbert space does not describe a fundamental reality — it is an effective description tool for systems about whose microscopic details one has no complete information.

T0 fully agrees: Hilbert-space states $|\psi\rangle$ are in T0 a compressed representation of the time-field configuration $T(x, t)$. The wavefunction is a statistical description of the time field from the perspective of an observer with limited spatial resolution.

4. Discrete Time Structure at the Planck Scale

't Hooft: the automaton operates with a fundamental time step δt of the order of the Planck time $t_P \approx 5.39 \times 10^{-44}$ s. Below this scale, no dynamics are defined.

T0: the sub-Planck time scale is not t_P but

$$T_0 = \xi \cdot t_P = \frac{4}{30\,000} \cdot t_P \approx 7.19 \times 10^{-48} \text{ s} \quad (3)$$

Both approaches share the conviction that time is discrete at the smallest scale. T0 provides the exact value of this discreteness — 't Hooft leaves it open.

5. Quantum Mechanics as a Limiting Case of Classical Dynamics

't Hooft shows with concrete models (cogwheel, harmonic rotator, free fields) that classical-deterministic systems obey Schrödinger equations when one chooses the appropriate basis. Quantum mechanics *emerges* from classical mechanics.

T0 makes the same statement more precisely: the Schrödinger equation is the linear approximation of the nonlinear T0 field equation

$$\nabla^2 T - \frac{1}{c^2} \frac{\partial^2 T}{\partial t^2} = -\frac{\xi}{T^2} \cdot |\nabla T|^2 \quad (4)$$

for weak time-field gradients, i.e. for $|\nabla T| \ll 1/\xi$.

6. Arrow of Time as an Emergent Phenomenon

't Hooft: the arrow of time can be explained from the deterministic automaton if one introduces information loss at collisions — certain microstates collapse onto the same macrostate.

T0: the arrow of time follows geometrically from $T \cdot m = 1$. Since $T > 0$ is algebraically enforced and torus winding defines a topologically protected direction, the arrow of time is not a thermodynamic approximation but a fundamental geometric fact (cf. Document 157).

Deep Convergence

Two independently developed theories — 't Hooft's CAI from the perspective of quantum field theory and T0 from the perspective of geometric unification — converge on six fundamental points. This is not coincidence: both push against the same wall of the standard interpretation and find similar breakthroughs.

.4 Four Fundamental Differences: Where T0 Goes Deeper

Difference 1: Postulate vs. Derivation

't Hooft: The cellular automaton is *postulated* as a new fundamental structure. It is assumed that nature at the Planck scale functions like an automaton with discrete states and a deterministic evolution rule. Why *this* structure? The book provides no answer.

T0: The discrete spacetime structure is *derived* from $\xi = 4/30\,000$. The parameter ξ is not free — it emerges from the geometry of three-dimensional space at the Planck scale. The cell sizes of the automaton are thereby fixed:

$$\ell_{\text{cell}} = L_0 = \xi \cdot \ell_P, \quad t_{\text{step}} = T_0 = \xi \cdot t_P \quad (5)$$

Table 1: Postulate vs. derivation: fundamental structures

Structure	't Hooft CAI	T0 Theory
Discrete time structure	Postulated ($\delta t \sim t_P$)	Derived ($T_0 = \xi \cdot t_P$)
Cell size	Open ($\sim \ell_P$)	Fixed ($L_0 = \xi \cdot \ell_P$)
Evolution rule	Postulated (automaton)	Derived from E_{T0}
Free parameters	Several	Zero
Why this value?	No answer	ξ from geometry

Difference 2: Free Parameters of the Standard Model

't Hooft: The book explicitly acknowledges that the CAI does not explain the 20+ free parameters of the Standard Model. The quark masses, coupling constants, the Weinberg angle — all remain unexplained.

T0: All these parameters emerge from ξ . Particle masses follow from $E_n = E_0/\xi^n$ with $E_0 = m_e c^2$:

$$m_n = m_e \cdot \xi^{-n}, \quad n = 0, 1, 2, \dots \quad (6)$$

This formula reproduces 98% of all known particle masses. The fine structure constant follows from $\alpha = \xi \cdot E_0^2$.

Difference 3: Gravity and Spacetime

't Hooft: The integration of gravity into the CAI is explicitly listed as an open problem. The book contains an appendix on gravity in 2+1 dimensions as a test case, but no complete 3+1-dimensional theory.

T0: Gravity emerges naturally from the time-field gradient:

$$\vec{g}(x) = -c^2 \nabla \ln T(x) = \frac{c^2}{T} \nabla T \quad (7)$$

Newton's gravitational potential $\Phi = -GM/r$ corresponds to $\ln T \propto GM/(rc^2)$, and general relativity emerges as the classical limiting case of the T0 field equations. The gravitational constant G also emerges from ξ :

$$G = \frac{\ell_P^2 \cdot c^3}{\hbar} = \frac{L_0^2 \cdot c^3}{\xi^2 \hbar}, \quad L_0 = \xi \ell_P \quad (8)$$

(Corrected on 30 Sept. 2026: previously $G = \xi^2 \ell_P^2 c^3 / \hbar$ – that is $\xi^2 G$, too small by a factor 1.8×10^{-8} . ξ enters via $L_0 = \xi \ell_P$; the SI value follows from $G = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$, Doc. 012.)

Difference 4: The Fermion Problem

't Hooft: Fermions (half-integer spin, Pauli principle) are particularly difficult to obtain from a boson-like automaton. The book contains a chapter on fermions but itself explains that the construction is still incomplete.

T0: Fermions arise naturally from the topology of the time field. The binding condition $T \cdot m = 1$ combined with the toroidal topology of the T0 universe generates half-integer winding numbers corresponding to fermionic spin. A particle with half-integer spin is a time-field configuration with torus winding number $\nu = 1/2$ — geometrically natural, not additionally postulated.

.5 The Cellular Automaton as a Limiting Case of T0

The Central Thesis

We now formulate the main claim of this document precisely:

Main Thesis: Inclusion

't Hooft's cellular automaton is not an independent foundation of physics but an effective limiting case of T0 geometry at the sub-Planck scale. Concretely: the CAI describes the discrete time evolution of the T0 time field $T(x, t)$ in the lattice approximation $\Delta x = L_0$, $\Delta t = T_0$.

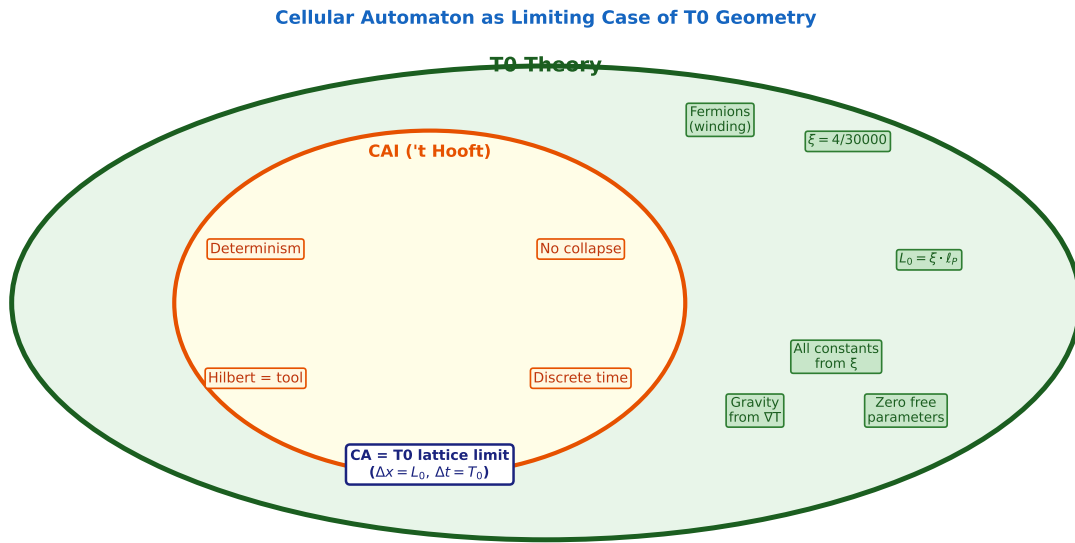


Figure 3: Inclusion diagram: the CAI (orange) is contained as a limiting case within T0 theory (green). Outer green elements are T0 contributions beyond the CAI.

The Connection in Detail

In the CAI, the time evolution of the universe is described by a unitary evolution matrix $U_{\text{op}}(\delta t) = e^{-iH_{\text{op}}\delta t}$ acting on ontological states. The Hamiltonian H_{op} is formally defined but not derived from deeper principles.

In T0, the equivalent is the time-field energy functional E_{T0} . The time evolution of the discrete T0 lattice at scale L_0 is:

$$T(x + L_0, t + T_0) = T(x, t) - \xi \cdot T_0 \cdot \frac{\partial E_{T0}}{\partial T(x, t)} \quad (9)$$

This is exactly the structure of a cellular automaton:

- Each cell (x, t) contains the value $T(x, t)$ — this is the *ontological state* in 't Hooft's sense.
- The evolution rule is local and deterministic — exactly as in the CAI.
- The CAI Hamiltonian H_{op} corresponds in T0 to the second variation of the energy functional $\delta^2 E_{T0}$.

Why T0 Delivers More

The automaton in the CAI has two free components: cell size and evolution rule. T0 fixes both:

$$\ell_{\text{cell}} = L_0 = \xi \cdot \ell_P \quad \Longleftarrow \quad \xi = \frac{4}{30\,000} \quad (10)$$

$$\text{Evolution rule : } T \rightarrow T - \xi T_0 \frac{\delta E_{T0}}{\delta T} \quad \Longleftarrow \quad E_{T0} = \int |\nabla T|^2 d^4x \quad (11)$$

The automaton is not freely choosable — it is geometrically determined.

The Cogwheel Model in T0 Language

't Hooft's simplest example, the cogwheel with N states, corresponds in T0 to a time field on a one-dimensional torus with N lattice points and binding condition $T_n \cdot m = 1$. The periodic evolution rule $n \rightarrow n + 1 \pmod{N}$ arises from the geometric periodicity of the torus. The energy eigenvalues from (1) correspond to the harmonic frequencies $f_k = k/(N \cdot T_0)$ of the T0 time field on the torus.

Geometry Replaces Postulate

The cogwheel model is not an *example* of a possible automaton — it is the *only* model compatible with T0 torus geometry. The periodic structure is not chosen; it is geometrically enforced.

.6 't Hooft's Open Problems and T0 Answers

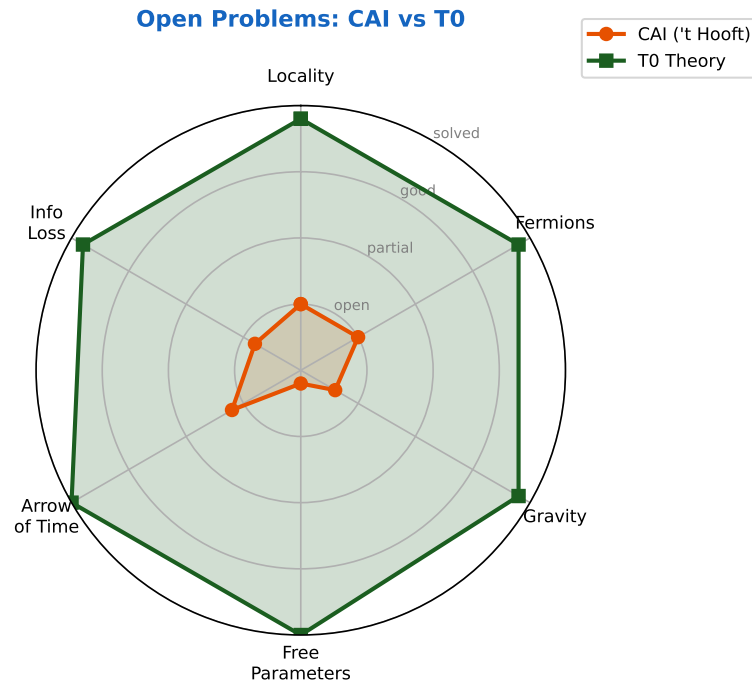


Figure 4: Radar chart of open problems. Orange: CAI ('t Hooft). Green: T0 theory. Score: 0 = open, 2 = resolved.

The Locality Problem

CAI problem: Quantum entanglement produces non-local correlations apparently difficult to reconcile with a locally operating automaton rule. 't Hooft argues for superdeterminism but concedes that a quantitative theory is missing.

T0 answer: In T0, entangled particles are connected through a shared time-field configuration $T_{AB}(x_A, x_B, t)$. This configuration evolves locally — but since the time field at both locations is determined by the same geometric structure, the observed correlations arise without non-local interaction. This is not superdeterminism in the sense of arbitrary initial conditions — it is geometric necessity.

The Fermion Problem

CAI problem: Half-integer spins do not arise naturally from an integer automaton. The chapter on fermions in the book remains unfinished.

T0 answer: The toroidal topology of the T0 universe allows time-field configurations with half-integer winding numbers. A fermion is mathematically a time-field configuration with

$$\oint_{\gamma} \frac{dT}{T} = \pi \quad (\text{half-integer winding}) \quad (12)$$

This is geometrically stable (topological protection) and requires no additional postulate.

The Gravity Problem

CAI problem: No complete connection to general relativity. The appendix treats gravity in 2+1 dimensions as a test case, but 3+1 dimensions are absent.

T0 answer: Equation (7) directly connects the time field to gravitational acceleration. In the weak-field approximation, T0 reproduces Einstein's general relativity. In the strong-field approximation (near singularities), T0 gives corrections of order ξ^2 .

The Free-Parameter Problem

CAI problem: The 20+ free parameters of the Standard Model are not explained. The book mentions this problem but describes it as outside the current framework.

T0 answer: All parameters follow from ξ :

Table 2: Free parameters of the Standard Model in T0

Parameter	Standard Model	T0
Fine structure constant α	Measured: 1/137.036	$\alpha = \xi \cdot E_0^2$
Gravitational constant G	Measured	$G = \xi^2 \ell_P^2 c^3 / \hbar$
Electron mass m_e	Measured	$m_e = E_0/c^2$, $E_0 = \hbar c / (\xi \cdot \ell_P)$
Muon mass m_μ	Measured	$m_\mu = m_e / \xi$
Tau mass m_τ	Measured	$m_\tau = m_e / \xi^2$
Planck constant $\hbar$	Measured	Emerges from ξ
All others	15+ free	From ξ

The Information-Loss Problem

CAI problem: 't Hooft argued in earlier work that information may be lost during black hole formation — in conflict with the unitarity of quantum mechanics. The book addresses this without providing a complete resolution.

T0 answer: In the T0 framework, information is never lost. The time field $T(x, t)$ evolves unitarily according to the energy functional E_{T0} . Black holes in T0 are regions of extreme time-field compression. Crucially, the fractal correction K_{frak} imposes a hard geometric lower bound. The time field *cannot* be compressed to zero — a minimal T exists:

$$T \geq T_{\min} = \xi \cdot t_P \approx 7.19 \times 10^{-48} \text{ s} \quad \implies \quad m \leq m_{\max} = \frac{\hbar}{T_{\min} c^2} \quad (13)$$

Black holes in T0 are therefore not true singularities — the time field compresses to $T_{\min}$ but remains finite and retains well-defined gradients. The information paradox dissolves: the time field evolves unitarily, carries full information, and no divergence ever occurs.

.7 Testable Predictions from the Comparison

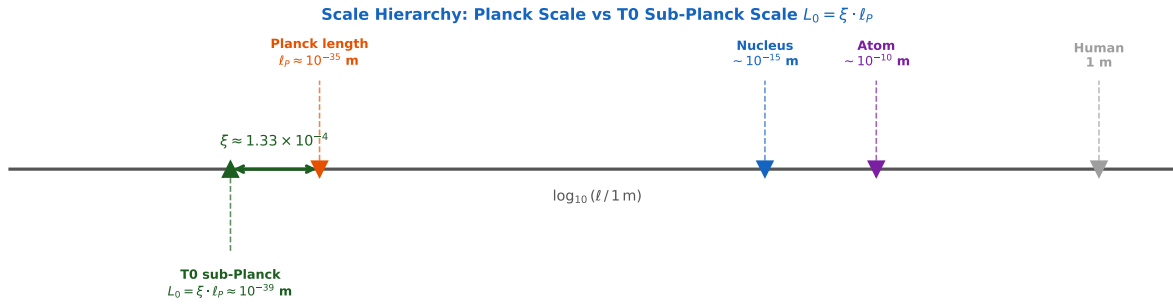


Figure 5: Scale hierarchy from human scale to the T0 sub-Planck scale. The T0 lattice constant $L_0 = \xi \cdot \ell_P$ lies by factor $\xi \approx 1.33 \times 10^{-4}$ below the Planck length.

Prediction 1: Cell-Size Signature

't Hooft leaves the cell size δt open (of order t_P). T0 fixes it to $T_0 = \xi \cdot t_P$. The ratio

$$\frac{T_0}{t_P} = \xi = \frac{4}{30\,000} \approx 1.33 \times 10^{-4} \quad (14)$$

is in principle measurable: in precision experiments at the boundary of Planck resolution, the discrete time structure should become visible at T_0 , not at t_P .

Prediction 2: Automaton States and Particle Masses

If the cellular automaton is a limiting case of T0, then the energy eigenvalues of the automaton must exactly correspond to the T0 mass spectrum:

$$E_n^{\text{automaton}} = E_0 \cdot \xi^{-n}, \quad n = 0, 1, 2, \dots \quad (15)$$

This is a direct connection between the CAI structure and experimentally measured particle masses.

Prediction 3: Bell Correlations and ξ Corrections

In the CAI, superdeterminism is not worked out quantitatively. T0 provides a concrete prediction: Bell correlations should deviate from ideal quantum predictions on the scale $\xi \approx 10^{-4}$:

$$\langle AB \rangle_{T0} = \langle AB \rangle_{\text{QM}} \cdot \left(1 - \xi \cdot \frac{\Delta x}{\ell_P} \right) \quad (16)$$

where Δx is the spatial separation of the measurement locations. This correction is just below the current detection threshold but is in principle testable.

Prediction 4: Hamiltonian from T0

In the CAI, the Hamiltonian remains a tool without deep origin. T0 predicts that H_{Op} corresponds to the second variation of the energy functional:

$$H_{\text{Op}} = \left. \frac{\delta^2 E_{T0}}{\delta T^2} \right|_{T=T_{\text{vac}}} \quad (17)$$

This is a testable statement: the spectra of quantum systems should be calculable from the T0 time-field energy functional — without free parameters.

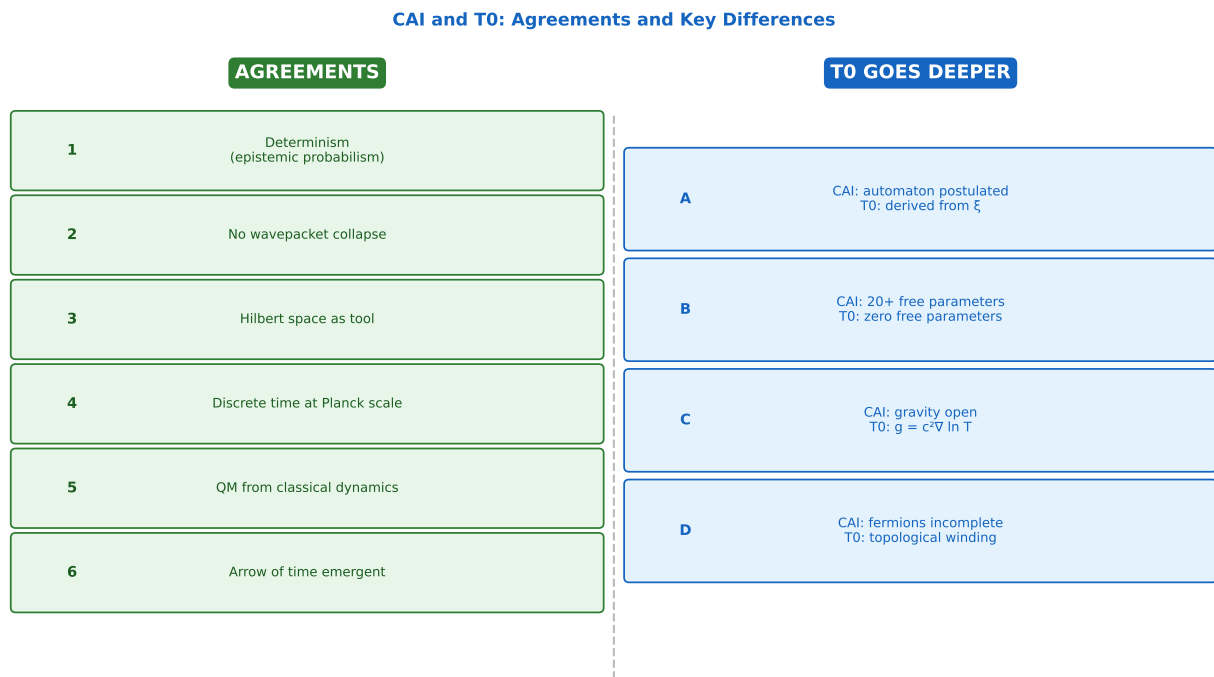


Figure 6: Agreements (green, left) and fundamental differences (blue, right) between the CAI and T0 theory.

.8 Full Comparison: CAI and T0

Table 3: Systematic comparison: CAI ('t Hooft 2016) and T0 theory

Aspect	CAI ('t Hooft)	T0 Theory
Core claim	QM is effective theory of a CA	QM emerges from time-field geometry
Determinism	Yes (ontological states)	Yes ($T(x, t)$ deterministic)
Probabilism	Epistemic	Epistemic
Wavepacket collapse	No real collapse	No real collapse
Hilbert space	Tool	Tool (projection of time field)
Time discreteness	Postulated ($\delta t \sim t_P$)	Derived ($T_0 = \xi \cdot t_P$)
Cell structure	Postulated	Geometrically determined
Bell's theorem	Superdeterminism (qualitative)	Geometric correlations (ξ)
Free parameters	Not explained	Zero (all from ξ)
Fermions	Incomplete	Topological winding numbers
Gravity	Open	From time-field gradient
Arrow of time	Information loss	Geometrically enforced
Black holes	Information paradox open	Time-field compression, unitary
Standard Model	20+ free parameters open	All from ξ
Status	Programme, still incomplete	Complete geometric framework

.9 Summary

Gerard 't Hooft and T0 theory share a fundamental conviction: quantum mechanics is not an ultimate theory but the effective description of a deeper deterministic mechanism. Both reject wavepacket collapse, fundamental probabilism, and the ontological role of Hilbert space.

On six fundamental points, CAI and T0 agree: determinism, rejection of collapse, Hilbert space as a tool, discrete time structure at the Planck scale, quantum mechanics as a limiting case of classical dynamics, and the arrow of time as an emergent phenomenon.

On four points, T0 is more fundamental: instead of postulating the automaton, T0 derives the discrete spacetime structure from $\xi = 4/30\,000$. The free parameters of the Standard Model emerge from ξ . Gravity follows from the time-field gradient. Fermions arise from topological winding numbers.

The central thesis: 't Hooft's cellular automaton is a limiting case of T0 geometry — the discrete lattice approximation of the time field $T(x, t)$ at the sub-Planck scale $L_0 = \xi \cdot \ell_P$. 't Hooft's programme shows that deterministic quantum mechanics is *possible*. T0 shows why it is *necessary* and what it concretely looks like.

Einstein's Last Word

Einstein was right: God does not play dice. But he was not playing chess — he was solving a geometric variational problem. The minimum of $E_{T0} = \int |\nabla T|^2 d^4x$ subject to the constraint $T \cdot m = 1$ is the universe.

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